

■ 1.5.11 Mathematics in *Mathematica* Packages

This section has described some of the many mathematical operations that are built into *Mathematica*. Many further mathematical operations are defined in external *Mathematica* packages, as discussed in Section 1.3.8.

Mathematica does not intrinsically know how to do Laplace transforms. This reads in a *Mathematica* package for doing Laplace transforms.

```
In[1]:= <<Laplace.m
```

Now *Mathematica* can do Laplace transforms.

```
In[2]:= Laplace[(1 + 2 t) Exp[-2 t], t, s]
```

$$\text{Out}[2]= \frac{2}{(2 + s)^2} + \frac{1}{2 + s}$$

You will have to look on your particular computer system to see what *Mathematica* packages are available. One point to be careful about is that if you ever use a name like **Laplace** *before* you read in the package that defines **Laplace**, then *Mathematica* may get confused about whether it should use your version of **Laplace**, or the one that was defined in the package. Page 385 discusses this issue in more detail.

Fourier.m	symbolic Fourier transforms and their inverses
Laplace.m	Laplace transforms and their inverses
ODE.m	functions for manipulating and solving ordinary differential equations
RungeKutta.m	numerical integration of differential equations
Trigonometry.m	trigonometric identities and simplification rules
VectorAnalysis.m	vector analysis

A few common *Mathematica* packages.